What is the origin of kinks in a shallow neural network?

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In a shallow neural network with ReLU activation, **kinks** arise as a direct consequence of how the hidden units behave.

Each neuron defines a piecewise linear function. The network's output is simply the sum of all of these, so the resulting function is also piecewise linear.

Let's see the output of a ReLU neuron:
A hidden neuron computes

$$h_i(x) = \max(0, \theta_i + \omega_i x),$$

where

- ω_i is the input weight,
- θ_i is the bias.

The ReLU function has two distinct regimes:

$$h_i(x) = \begin{cases} 0, & \theta_i + \omega_i x < 0, \\ \theta_i + \omega_i x, & \theta_i + \omega_i x \geq 0. \end{cases}$$

That is, before a certain point the output stays constant, and after it, the output becomes a linear function.

The transition occurs when the argument of the ReLU changes sign.
Solving

$$\theta_i + \omega_i x = 0,$$

we get

$$x = -\frac{\theta_i}{\omega_i}.$$

This value is the **kink** generated by the neuron. At this point, the slope of the function changes.

If the hidden layer contains three neurons, the network's output is

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3,$$

where

- ϕ_0 is the output bias,
- ϕ_i are the weights connecting the hidden layer to the output.

Each term $\phi_i h_i(x)$ is a piecewise linear function with a single kink. Summing all these functions gives another piecewise linear function.

Let's see why several kinks appear.
Each neuron has its own point

$$x_i = -\frac{\theta_i}{\omega_i}.$$

When the input value reaches one of these points, a neuron switches state (from inactive to active, or vice versa).

As a result:

- that neuron's contribution changes,
- the overall slope of the output changes,
- a new kink appears in the function represented by the network.

Because of this, a network with several neurons can exhibit multiple slope changes.

Before the kink, it contributes a line (or a constant value). After the kink, it contributes another line with a different slope.

The neural network builds complex functions simply by summing many of these hinges.